Keypad Based Product Counting and Automatic Batch Packing System with Station Sensing using LPC2129

Embedded Systems Project Report

1. Abstract

This project uses an LPC2129 (ARM7) microcontroller with two IR sensors and a keypad to count products moving on a conveyor belt driven by a DC motor. The operator sets the required batch quantity through the keypad and presses the start key. The first IR sensor is placed at the start of the belt and increases the product count through an external interrupt every time a product passes. The second IR sensor is placed at the packing station at the end of the belt and detects whether a product has reached the station. When the batch quantity is reached, the motor stops, a buzzer beeps and the LCD shows the final count. If the first sensor counts a product but the second sensor does not detect it at the packing station within a set time, the system treats it as a missing product (fallen or blocked on the belt), stops the motor and shows a fault message on the LCD. A red LED glows while the belt is running and a green LED glows when the batch is complete. The system reduces manual counting errors and detects belt faults in packaging and production lines.

2. Block Diagram

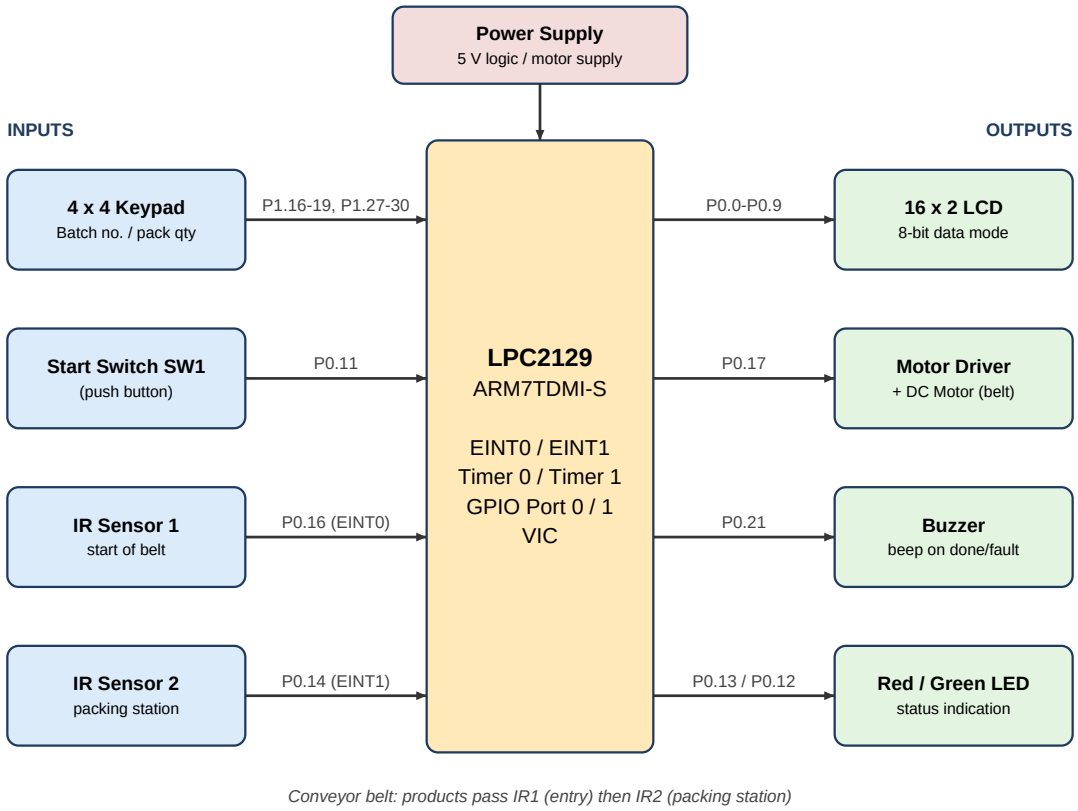

Fig. 1: Block diagram of the keypad based product counting and batch packing system

3. Components Used

3.1 Hardware

S.No	Component	Purpose in the project
1	LPC2129 (ARM7TDMI-S) microcontroller board	Main controller. Handles keypad scanning, two external interrupts, timer-based timeout, LCD, motor, buzzer and LED control.
2	IR sensor module (2 nos.)	IR1 at the start of the belt (EINT0) detects a product entering. IR2 at the packing station (EINT1) confirms arrival. Both trigger on a falling edge.
3	4 x 4 matrix keypad	Operator enters the batch number and the number of packs for the batch.
4	16 x 2 character LCD (HD44780 compatible)	Displays prompts, batch number, running count (count/target), completion and fault messages. Used in 8-bit mode.
5	DC motor with driver (transistor / relay / motor driver IC)	Drives the conveyor belt. Switched on and off by one GPIO pin.
6	Conveyor belt setup	Carries the products from the entry point to the packing station.
7	Buzzer	Audible indication when the batch is complete or a fault occurs.
8	Red and green LEDs with current limiting resistors	Visual status indication of running and batch-complete states.
9	Push button switch (SW1)	Starts the system and the batch entry sequence.
10	Regulated power supply	5 V for logic and sensors; supply for the motor as required.

3.2 Software

Embedded C program (main application, `lcdheader.h` for LCD routines and `keypad.h` for keypad scanning) written for the LPC21xx family, typically built with Keil uVision and loaded through the UART bootloader.

3.3 Pin Allocation

Pin(s)	Connected to	Direction / Function
P0.0 - P0.7	LCD data lines D0-D7	Output
P0.8, P0.9	LCD RS and E	Output
P0.11	Start switch SW1	Input, active low
P0.12	Green LED	Output
P0.13	Red LED	Output
P0.14	IR sensor 2 (packing station)	EINT1, falling edge interrupt
P0.16	IR sensor 1 (start of belt)	EINT0, falling edge interrupt
P0.17	Motor driver input	Output
P0.21	Buzzer	Output
P1.16 - P1.19	Keypad columns	Input
P1.27 - P1.30	Keypad rows	Output (row scan)

4. Description

4.1 Working Principle

1. **Initialisation.** The pins are configured, the LCD is initialised and the two external interrupts (EINT0 and EINT1) are set as edge triggered on the falling edge and assigned to vectored IRQ slots in the VIC. The LCD shows *PRODUCT COUNTING SYSTEM* followed by *PRESS THE SWITCH TO ON*.
2. **Start and data entry.** When switch SW1 is pressed, the LCD asks for *BATCH NO* and *NO OF PACKS*. Each value is read from the 4 x 4 keypad by the row-column scanning routine (keys return values 1 to 16) and echoed on the LCD.
3. **Belt start.** The LCD shows *MACHINE STARTED*, the motor is switched on and the display changes to *WAITING FOR PRODUCTS*.
4. **Product enters (IR1).** A product cutting the beam of the first IR sensor generates EINT0. Its interrupt service routine sets a flag, resets and starts Timer 1 and requests a display update. Timer 1 is prescaled to count in 1 second steps (PR = 15,000,000 - 1 for a 15 MHz peripheral clock).
5. **Product reaches the packing station (IR2).** The second IR sensor generates EINT1. If the product arrives within the allowed time (Timer 1 value of 5 s or less), the pack count is incremented, Timer 1 is stopped and the LCD shows the running status as *count/target*.
6. **Fault detection.** If Timer 1 exceeds 5 s without IR2 detecting the product, it is treated as missing (fallen or blocked on the belt). The motor is stopped, the buzzer is operated and the LCD shows *PACK NOT TRACED / RESTART*. The system returns to waiting for the start switch.
7. **Batch completion.** When the count reaches the number of packs entered, the motor is stopped, the buzzer is operated and the LCD displays *BATCH : n COMPLETED* for about 5 s. The system then returns to the start prompt, ready for the next batch.

4.2 Key Features

- Interrupt driven sensing, so no product passing a sensor is missed while the LCD is being updated.
- Timer based supervision of the travel time between the two sensors to detect missing or jammed products.
- Operator-set batch number and batch quantity through the keypad.
- Automatic motor stop with audible and visual indication at batch completion or fault.
- Reduces manual counting errors in packaging and production lines.

4.3 Applications

Packaging lines, bottling and pharmaceutical tablet or carton counting, small scale production units and any conveyor based batch counting system.